Exact Rank Transitions through p=32 and a Half-Integral Optimum at p=53

Inverse Hilbert-Schmidt self-commutator optimization on the ray $F_{(p,2p+1)}$, with exact rational hive, dual, and Farkas certificates

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STATUS. Completed manuscript. Exact costs and least attaining ranks are certified for every $4 \le p \le 32$ on $q=2p+1$, proving rank transitions at p=8, 15, 27, and 28 and a cost-slope transition at p=29. An additional exact theorem at p=53 identifies a half-integral optimum and refutes the old integer candidate. No all-p theorem is claimed.

Abstract

For the traceless Hermitian target $F_{(p,q)}=\mathrm{diag}(q$ repeated p times, $-p$ repeated q times), let kappa(F) be one half of the least squared Hilbert-Schmidt norm of a factor C satisfying $CC^* - C^*C = 2F$, and let $r_*(F)$ be the least rank among minimizers. On $q=2p+1$ we certify the complete finite frontier for $4 \le p \le 32$. The minimum-rank excess r_*-q equals 0 on $4 \le p \le 7$, 1 on $8 \le p \le 14$, 2 on $15 \le p \le 26$, 3 at p=27, and 4 on $28 \le p \le 32$. Thus p=27 and p=28 are two consecutive, rigorously separated rank transitions. Writing $M_p=\kappa(F_{(p,2p+1)})-(3p^2+2p)$, the exact data also prove $M_p=6p-30$ for $15 \le p \le 28$ and $M_p=7p-58$ for $29 \le p \le 32$, so the cost slope changes at p=29 rather than at the rank transition. Separately, $\kappa(F_{(53,107)})=8847$ and $r_*=115$. Its optimum is half-integral; the earlier integer candidate of trace 8843 is refuted by an integral Farkas certificate. Every new assertion is replayed in exact rational arithmetic.

1. Problem and spectral reduction

For a traceless Hermitian matrix F, define

$$\kappa(F) = (1/2) \min \{ \|C\|_{HS}^2 : CC^* - C^*C = 2F \}, \quad r_*(F) = \min \{ \mathrm{rank}(C) : C \text{ attains } \kappa(F) \}.$$

Set $P=(1/2)CC^*$ and $Q=(1/2)C^*C$. Then P and Q are positive and isospectral, $P-Q=F$, $\mathrm{tr}(P)=(1/2)\|C\|_{HS}^2=\kappa(F)$, and $\mathrm{rank}(P)=\mathrm{rank}(C)$. Conversely, positive isospectral P,Q with P-Q=F determine such a factor C by the polar construction. Thus the inverse problem is exactly a minimum-trace problem for positive isospectral matrices. At an optimum their common least eigenvalue is zero. The possible common spectra are governed by the Horn-Klyachko inequalities, equivalently by honeycombs or hives. A rank cap fixes the tail of this scaled spectrum to zero, turning both kappa and r_* into linear optimization questions over nested Horn faces [1-4].

2. Exact finite frontier

For every row below, a feasible primal spectrum and hive attain the displayed cost, an unrestricted rational dual matches it, and a second dual on the rank-at-most-(r_*-1) face has strictly larger value. The first block was the original exact frontier; the second block is the new exact closure.

p	q	kappa	r_*	r_*-q	status
4	9	65	9	0	exact
5	11	98	11	0	exact
6	13	137	13	0	exact
7	15	182	15	0	exact
8	17	233	18	1	exact
9	19	291	20	1	exact
10	21	355	22	1	exact
11	23	425	24	1	exact
12	25	501	26	1	exact
13	27	583	28	1	exact
14	29	671	30	1	exact
15	31	765	33	2	exact
16	33	866	35	2	exact
17	35	973	37	2	exact
18	37	1086	39	2	exact
19	39	1205	41	2	exact
20	41	1330	43	2	exact

On 4<=p<=20 the certified rank excess is 0, 1, and 2 on the three indicated intervals. The continuation below is proved by newly frozen primal-dual objects, not by extrapolation.

The new exact closure, 21<=p<=32

p	q	kappa	r_*	r_*-q	rank <= r_*-1 value
21	43	1461	45	2	1462
22	45	1598	47	2	1599
23	47	1741	49	2	1742
24	49	1890	51	2	1891
25	51	2045	53	2	2047
26	53	2206	55	2	2209
27	55	2373	58	3	2374
28	57	2546	61	4	2547
29	59	2726	63	4	2727
30	61	2912	65	4	2913
31	63	3104	67	4	3105
32	65	3302	69	4	3303

THEOREM 2.1 (finite phase diagram). On q=2p+1 and 4<=p<=32, the exact minimum-rank excess is 0 for 4<=p<=7, 1 for 8<=p<=14, 2 for 15<=p<=26, 3 for p=27, and 4 for 28<=p<=32. Hence the exact rank transitions in this range are p=8, 15, 27, and 28.

The cost and rank transitions do not coincide. Define M_p=kappa(F_(p,2p+1))-(3p^2+2p). Exact subtraction gives M_p=6p-30 on 15<=p<=28 and M_p=7p-58 on 29<=p<=32. Thus the cost correction changes slope at p=29, one step after the consecutive rank jumps at p=27 and p=28.

Candidate state suggested by the exact spectra

Every selected minimum-rank spectrum in the exact range admits a block decomposition

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s_p = (2p+A_p, 2p, p+B_p, C_p, 0^(p-|C_p|)).
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At the selected parameters E_m=3*2^m+m+1, the blocks suggested explicit integer partitions A_m, B_m, and C_m. This formula reproduces the first three transition parameters and predicted the frozen p=28 spectrum before its exact certificates were constructed. The complete exact closure now proves that p=27 and p=28 are the actual consecutive rank transitions. The selected parameter p=53 remains an out-of-sample falsifier of the stronger integer-spectrum claim.

m	p=E_m	final kappa	final r_*	final status
0	4	65	9	exact feasible
1	8	233	18	exact feasible
2	15	765	33	exact feasible
3	28	2546	61	exact optimum
4	53	8847	115	exact; old state NO-GO

3. Exact cost and rank at p=28

THEOREM 3.1. For F_(28,57)=diag(57 repeated 28 times, -28 repeated 57 times), kappa(F_(28,57))=2546. There exists an optimum of rank 61, and no optimum has smaller rank.

Upper certificate

The common positive spectrum is recorded below in run-length notation; a^b means b repetitions of a.

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(84, 71, 64^2, 60^4, 58^7, 57^13, 56, 36, 32^2, 30^3, 29^6, 28^16, 4, 2, 1^2,
0^24).
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Its trace is 2546 and its rank is 61. The frozen Littlewood-Richardson tableau has 85 rows and 2,380 cells in the required skew shape, with content (85 repeated 28 times). The replay checks the complete shape, weak row order, strict column order, exact content, and every prefix lattice inequality. Hence the spectrum is Horn feasible and gives kappa<=2546 at rank 61.

Matching lower certificates

An exact unrestricted hive dual has value 2546. A second exact dual on the rank-at-most-60 face has value 2547. Both use only integral or half-integral multipliers. Their supports are 3,126 and 3,210 rows. The first excludes any value below 2546; the second excludes every rank at most 60 at the optimum. Together with the LR construction, they prove Theorem 3.1.

Certificate logic

Object	Exact statement	Consequence
LR tableau	feasible, trace 2546, rank 61	kappa<=2546 and r_*<=61
unrestricted dual	dual value 2546	kappa>=2546
rank-60 dual	dual value 2547	no optimum has rank<=60

Combined with the exact p=27 predecessor, Theorem 2.1 establishes that p=28 is a genuine rank transition. It does not imply a formula for arbitrary p.

4. Exact cost and rank at p=53

THEOREM 4.1. kappa(F_(53,107))=8847, and the minimum attaining rank is 115.

Here F_(53,107)=diag(107 repeated 53 times, -53 repeated 107 times). Equivalently, there exists an optimum of rank 115 and none has smaller rank.

Exact upper certificate

A common spectrum of dimension 160, trace 8847, and rank 115 is given in run-length notation by

(159, 134, 243/2, 121, 229/2, 114^3, (221/2)^2, 110^5, 217/2, 108^12, 215/2, 107^24, 106, 137/2, 123/2, 61, 115/2, 57^2, 55^6, 109/2, 54^10, 107/2, 53^29, 17/2, 4, 2^2, 1^4, 0^45).

The frozen rational hive has 13,200 coordinates and denominator at most two. Exact replay checks all 481 boundary equalities and 38,319 hive, order, and nonnegativity inequalities; 34,589 inequalities are tight. This supplies kappa<=8847 at rank 115.

Matching lower certificates

An unrestricted exact hive dual has value 8847 and support 10,837. A second exact dual on the rank-at-most-114 face has value 8848 and support 10,949. Every stored multiplier is integral or half-integral. Stationarity is checked on all 13,200 coordinates. The first dual matches the upper construction; the second proves the strict predecessor-rank gap. Therefore the least rank among all norm minimizers is exactly 115.

Object	Exact statement	Consequence
rational hive	trace 8847, rank 115	kappa<=8847 and r_*<=115
unrestricted dual	dual value 8847	kappa>=8847
rank-114 dual	dual value 8848	no optimum has rank<=114

Why the old candidate state still matters

The previous integer recursion predicted a different rank-115 spectrum of trace 8843. Substitution leaves 13,041 free hive nodes. A 636-row integral Farkas combination cancels every free coefficient while its right side equals one, giving the contradiction 0 >= 1. Thus the old spectrum and cost prediction are false, even though the minimum-rank prediction 115 survives through a new half-integral Horn face.

Why a half-integral optimum is structurally possible

The saturation theorem concerns integral boundary triples after dilation; it does not say that every vertex obtained by allowing the common spectrum itself to vary is integral. Here the optimization projects the hive cone onto one shared boundary, intersects it with positivity and a rank face, and minimizes trace. Such a projected rational polyhedron may have fractional exposed vertices. At p=53 the denominator-two primal and denominator-two matching dual identify the same exposed face, while the integral trace-8843 candidate is separated by an integral Farkas certificate. Thus half-integrality is a property of the optimum face, not a numerical artifact.

5. Reproducibility and independent replay

The discovery path and the proof replay are deliberately separated. HiGHS and SciPy were used to locate primal spectra, tableaux, active rows, and a normalized Farkas combination. The frozen proof objects are then checked by scripts that use only the Python standard library and exact Fraction arithmetic.

Replay	Checks	Result
verify_lr_frontier_bundle.py	original exact frontier and p=28 LR objects	PASS
verify_p28_exact_duals.py	full and rank-60 dual stationarity/objectives	PASS
verify_exact_frontier_p21_p32.py	12 primals and 24 duals; exact phase diagram	PASS
verify_p53_exact_endpoint.py	hive plus full/rank-114 duals; 13,200 coordinates	PASS
verify_p53_independent.py	independent row/index reconstruction	PASS
verify_p53_endpoint_nogo.py	636 multipliers; 13,041 cancellations; RHS=1	PASS
python -O verifier	optimized execution must be refused	REFUSED

Canonical certificate identifiers

p=28 unrestricted dual: 72357c66fb80a75b5a28f221b2bc172b62dcefbbaccdc7131d5fdb257860646b

p=28 rank-60 dual: f1333e9fd9bb73c6faf05cf763c07800d93da1b8d5e6f55b28ac0fad0ef7e0ae

p=53 Farkas certificate: 2968b7c01103fb70bc3d0c57138b098f7995584a9b13b40f684df7627204367d

p=53 rank-115 primal: 1a85bbed330ac8115c3a817ee25b0d95a6314363c1ab79c0e11dd61617ad7dfd

p=53 unrestricted dual: a32442334b45bf11fdb00cd7e826399e19dca0fdaeb3ab95319f899f7cbb9d66

p=53 rank-114 dual: 4a1b24dffdf8a7a394a2c9395155887af75e86a82f192d86309c1bcda88d54de

Proof boundary

The replay is conditional on the classical Horn-Klyachko theorem and its hive model. It is not a proof-assistant formalization. Independent peer review and an exhaustive priority search have not been performed. No numerical solution is used as a final proof object.

6. Consequences and remaining boundary

The complete frontier establishes two consecutive rank transitions at p=27 and p=28, followed by a distinct cost-slope transition at p=29. The p=53 theorem raises the exact excess to eight: 115-107=8. More importantly, it separates three notions that the original recursion conflated. The predicted integer spectrum fails, its cost 8843 fails, but its rank 115 survives through a different half-integral optimum. A long exact fit can therefore identify a rank transition while missing the supporting Horn face and objective value.

> FINAL CLAIM BOUNDARY. Exact theorems: the full cost-and-rank phase diagram for 4<=p<=32, including rank transitions at 8, 15, 27, 28 and the cost-slope change at 29; and kappa(F_(53,107))=8847 with r_*=115. Exact negative result: the old p=53 integer spectrum is infeasible. Open: every all-p recurrence and a classification of the optimal Horn faces between or beyond these parameters.

References

[1] A. Horn, Eigenvalues of sums of Hermitian matrices, Pacific Journal of Mathematics 12 (1962), 225-241. DOI: 10.2140/pjm.1962.12.225.

[2] A. A. Klyachko, Stable bundles, representation theory and Hermitian operators, Selecta Mathematica 4 (1998), 419-445. DOI: 10.1007/s000290050037.

[3] A. Knutson and T. Tao, The honeycomb model of GL(n) tensor products I: Proof of the saturation conjecture, Journal of the American Mathematical Society 12 (1999), 1055-1090. arXiv:math/9807160.

[4] W. Fulton, Eigenvalues, invariant factors, highest weights, and Schubert calculus, Bulletin of the American Mathematical Society 37 (2000), 209-249. DOI: 10.1090/S0273-0979-00-00865-X.

[5] S. Fomin, W. Fulton, C.-K. Li, and Y.-T. Poon, Eigenvalues, singular values, and Littlewood-Richardson coefficients, arXiv:math/0301307.

Related exact result

L. Eriksson, Optimal-rank onset for inverse self-commutators, ARR-2026-5QQF95VHTC9GABH8. The present manuscript does not alter that archived record.

Authorship, computational assistance, and conflicts

Lluis Eriksson is the author and is responsible for the claims and proof boundary. Generative AI assisted corpus exploration, conjecture testing, code review, and drafting under author direction. Floating-point optimization was used only during discovery; every claim stated as exact has an exact replay. No external peer review or model evaluation is claimed. No conflict of interest is declared.

Scope and limitations

- Complete in scope for the exact 4<=p<=32 phase diagram and the p=53 theorem.
- No claim of a uniform p>=4 recurrence.
- No classification of all minimizers or of every p=53 optimal spectrum.
- Archive metadata and model assessments are not claims of peer review.